CE205 Databases and Information Retrieval

Class 3

Objective: To convert and check a relation into a desired normal form.

Example: The relation R (A, B, C) is decomposed into two relations R1(A, B) and R2(B, C). Find the natural join between them.

R(A, B, C)

A	B	C
1	2	3
1	2	5
4	2	3
4	2	5

R1 (A, B)

A	B
1	2
4	2

R2 (B, C)

B	C
2	3
2	5

- See the common attribute in both relations (which is B in the given relations)
- For a value of B in R1, see all corresponding rows in R2. Repeat it for all values of B. Resultantly, you will get

A	B	C
1	2	3
1	2	5
4	2	3
4	2	5

The result of the natural join equals the original relation R, so the decomposition is lossless.

Problem 1:

The relation ***PropertyItemSupplier*** is decomposed into the sub-relations ***PropertyItem***, ***ItemSupplier***, and ***PropertySupplier***.

PropertyItemSupplier (Legal state)

propertyNo	itemDescription	supplierNo
PG4	Bed	S1
PG4	Chair	S2
PG16	Bed	S2
PG4	Bed	S2

PropertyItem

propertyNo	itemDescription
PG4	Bed
PG4	Chair
PG16	Bed

ItemSupplier

itemDescription	supplierNo
Bed	S1
Chair	S2
Bed	S2

PropertySupplier

propertyNo	supplierNo
PG4	S1
PG4	S2
PG16	S2

1) Perform natural join between ***PropertyItem*** and ***ItemSupplier, i.e., PropertyItem*** $\bowtie$ ***ItemSupplier***
2) Perform natural join between ***ItemSupplier*** and ***PropertySupplier, i.e., ItemSupplier***, $\bowtie$ ***PropertySupplier***
3) Perform natural join between ***PropertyItem*** and ***PropertySupplier***, ***i.e., PropertyItem***, $\bowtie$ ***PropertySupplier***
4) Perform the natural join between the **result of Part (1)** and ***PropertySupplier.***

In all parts (1-4), determine whether the result is Lossless or Lossy Join-Decomposition.

Problem 2:

The following relation is given:

HOSTEL (StudentNumber, StudentName, Dorm, DormCost, Club, ClubCost, Trainer)

Assume that students pay different dorm costs depending on their room type. All club members pay the exact cost for the club. Each club may have multiple trainers; however, a student will have a single trainer for each club.

1) Add sample data in Hostel.
2) Find all functional dependencies in HOSTEL.
3) Find candidate key(s) in Hostel.
4) Normalise the relation to BCNF (directly or first normalise to 2nd and 3rd normal form). Preferably, use both approaches one by one.
5) State each sub-relation's primary key and foreign key.

Solution:

StudentNumber	StudentName	Dorm	DormCost	Club	ClubCost	Trainer